



FLIGHT DELAYS

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30 JANUARY, 2026



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INTRODUCTION

DATASET

U.S. DOT – BUREAU OF TRANSPORTATION
STATISTICS (BTS)

SCOPE

NOVEMBER 2024 -
OCTOBER 2025



THIS RESEARCH EMPLOYS **SUPERVISED MACHINE LEARNING**
TO QUANTIFY THE **INFLUENCE** OF **TEMPORAL** AND
OPERATIONAL CONSTRAINTS ON AVIATION PERFORMANCE.

THROUGH THE IDENTIFICATION OF CRITICAL DRIVERS—
INCLUDING **DEPARTURE TIME**, **AIRPORT PATTERNS** AND
ROUTES—THE PROJECT ESTABLISHES A PREDICTIVE MODEL
FOR **FLIGHT DELAYS**.

OBJECTIVES

IDENTIFY THE KEY FACTORS THAT CAUSE FLIGHT DELAYS, SUCH AS:

- TIME OF DAY,
- DAY OF WEEK
- MONTH → SEASON
- ROUTE

FLIGHT STATUS PREDICTION

BUILD A DEPLOYABLE FLIGHT STATUS PREDICTOR THAT ESTIMATES THE PROBABILITY A FLIGHT WILL BE >15 MINUTES DELAYED.



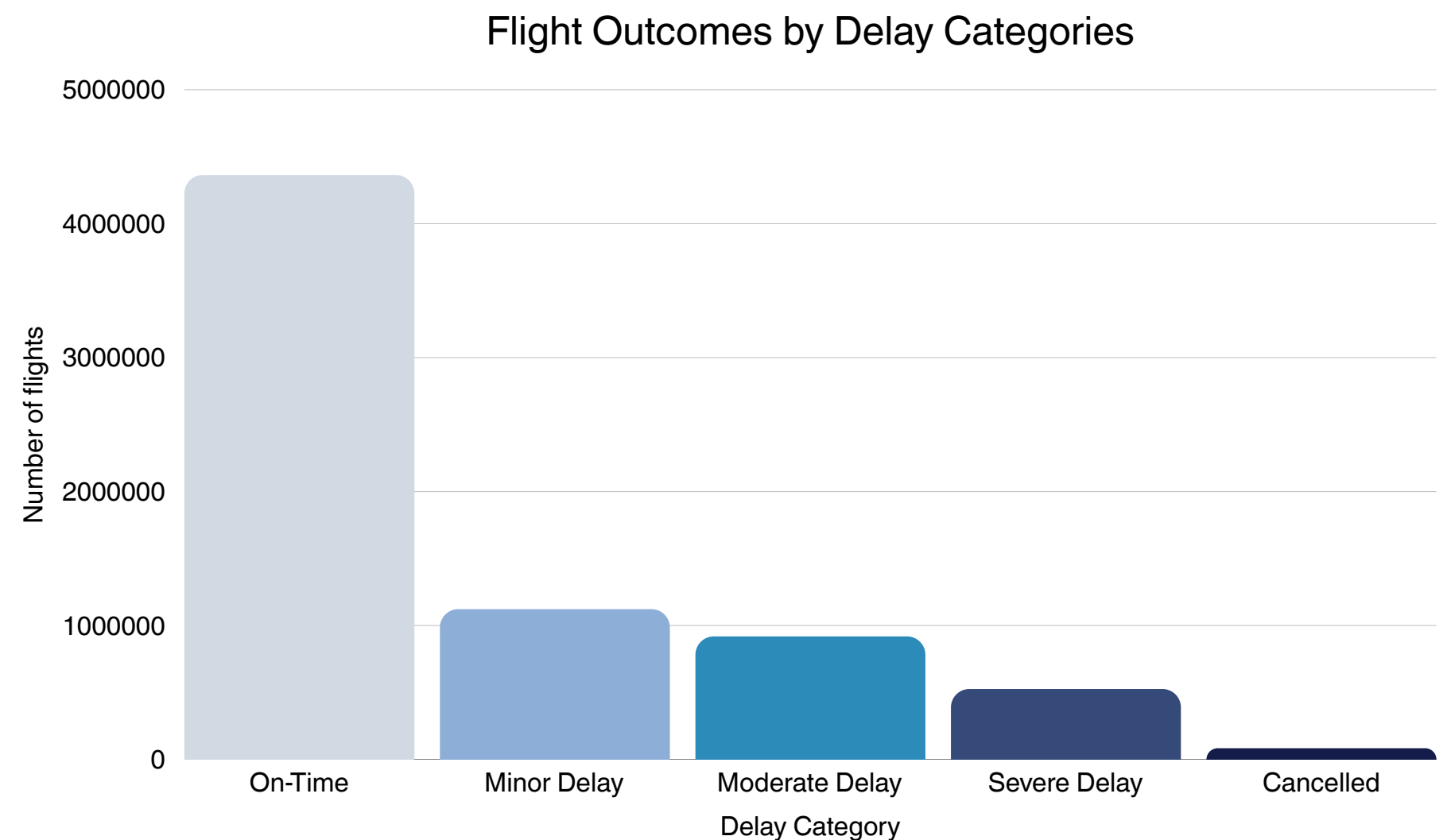
ANALYSIS *AND* INSIGHTS



EXPLORATORY DATA ANALYSIS

- Minor delay: <15 minutes
- Moderate delay: between 15 and 60 minutes
- Severe delay: >60 minutes

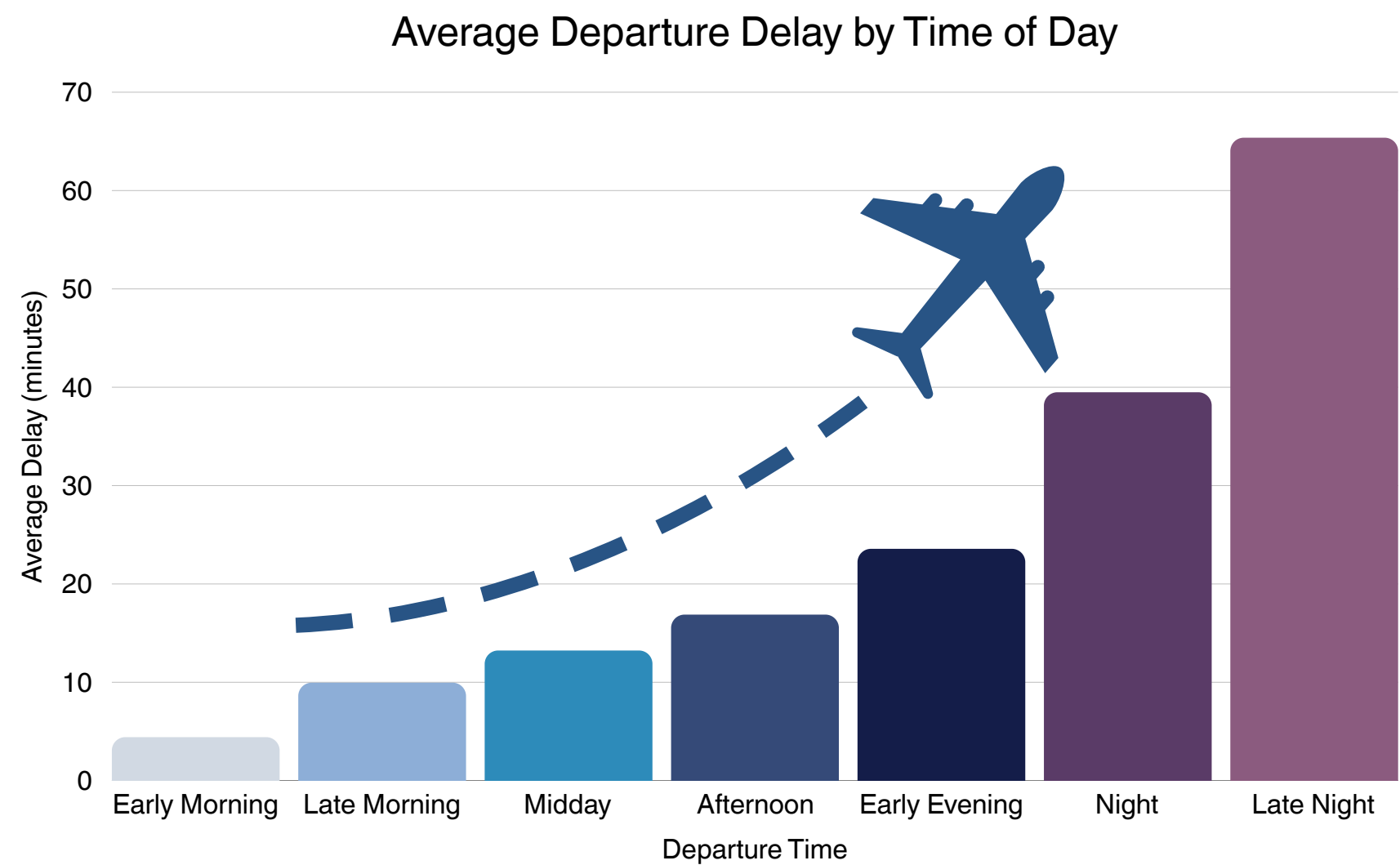
We see that the majority of flights are On-Time, so this will cause an Imbalance in our Machine Learning models.



TEMPORAL PERFORMANCE ANALYSIS I

- Late Night: 00:00 – 04:59
- Early Morning: 05:00 – 07:59
- Late Morning: 08:00 – 10:59
- Midday: 11:00 – 13:59
- Afternoon: 14:00 – 17:59
- Early Evening: 18:00 – 20:59
- Night: 21:00 – 23:59

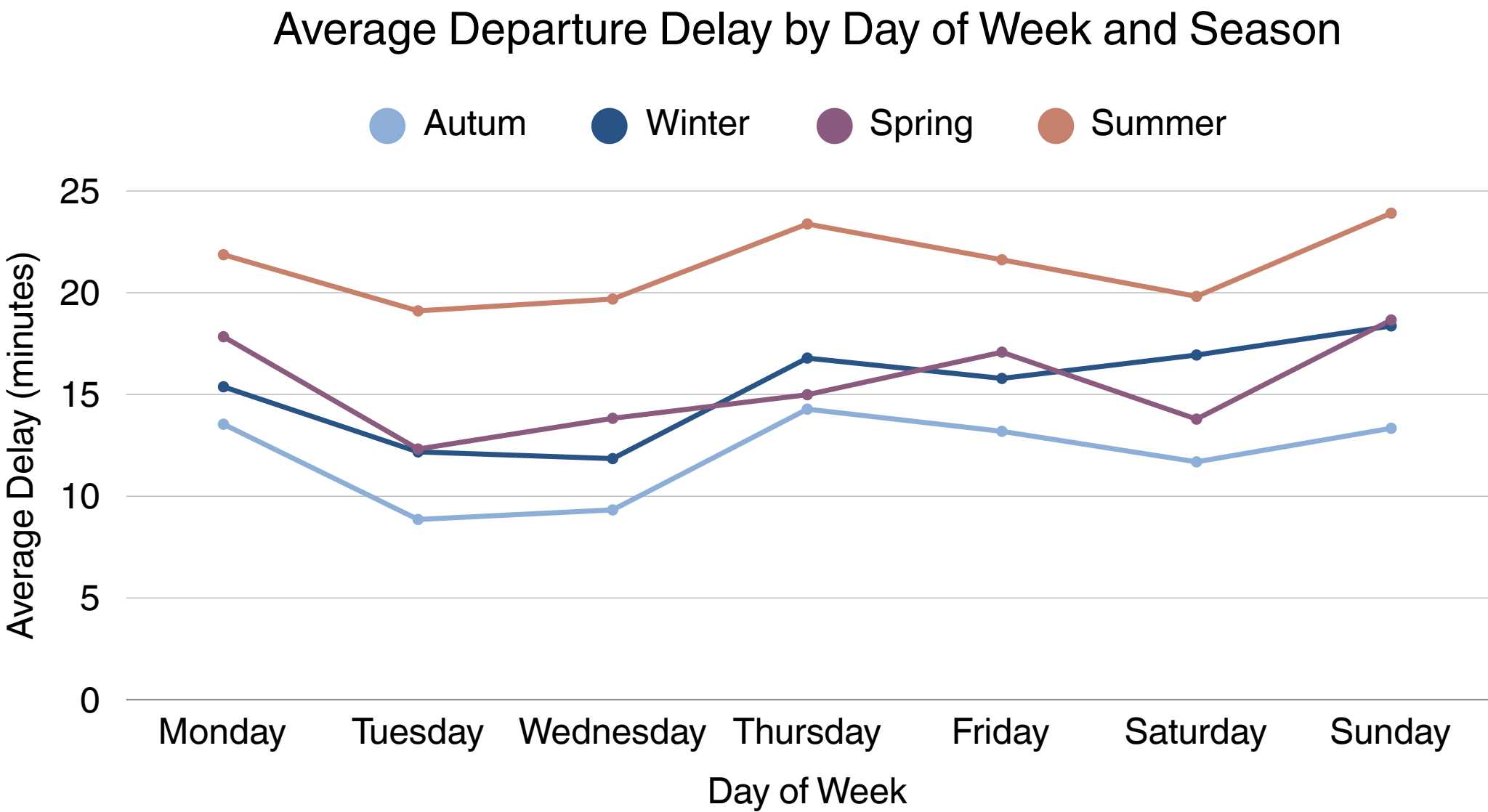
Conclusion: The likelihood of flight delays increases as departure times move later into the day.



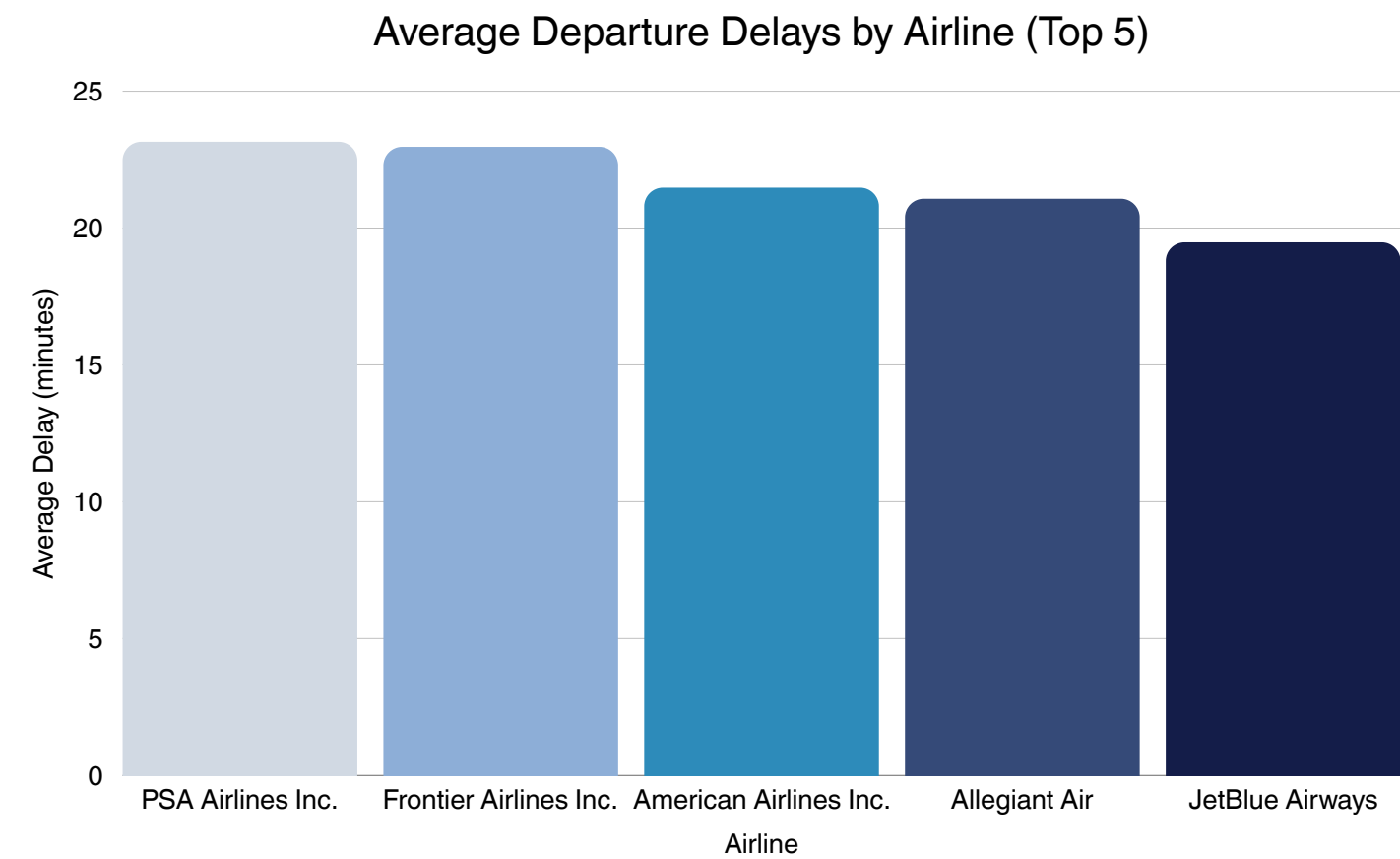
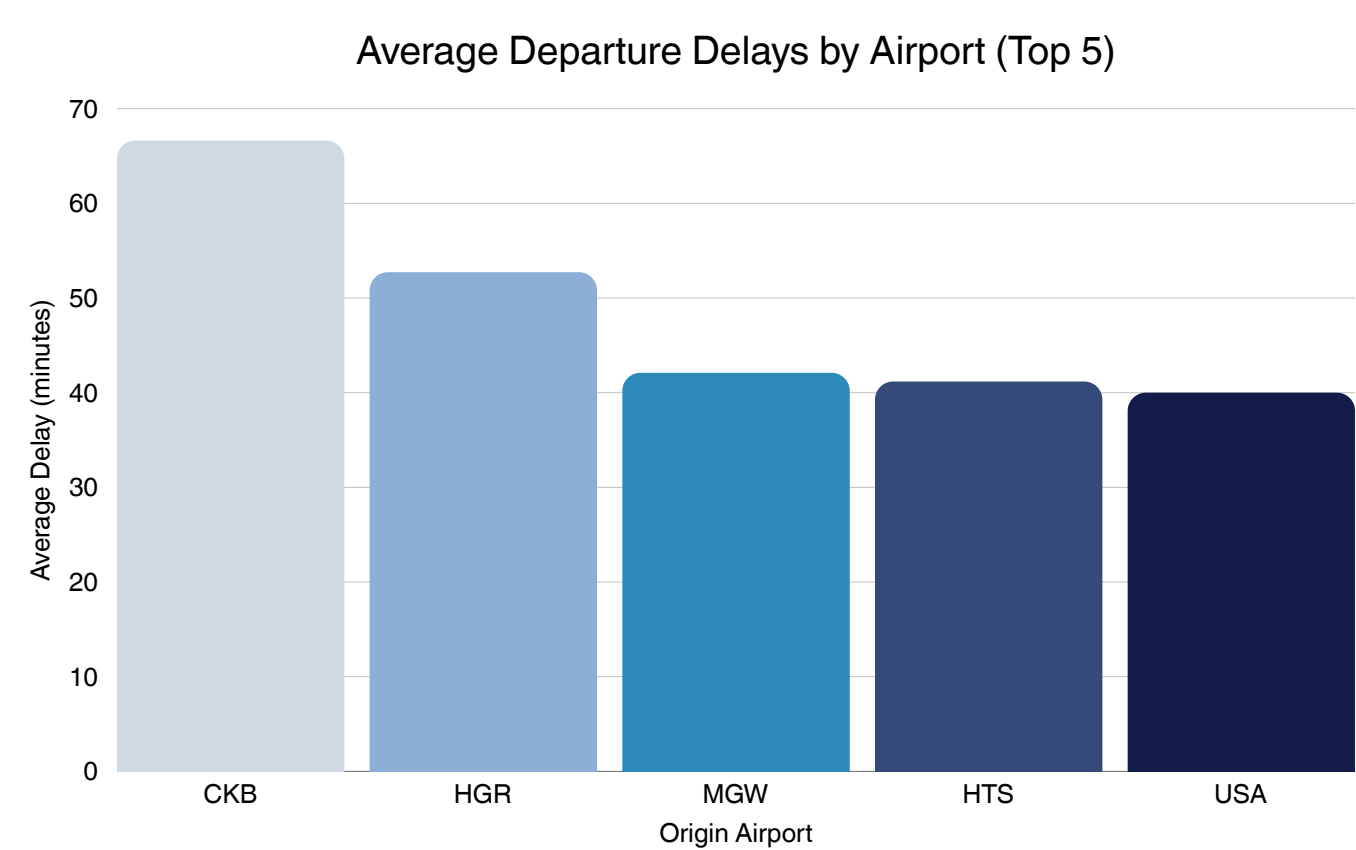
TEMPORAL PERFORMANCE ANALYSIS II

- Autum: Septembere, OCTpber and November
- Winter: December, January and February
- Spring: March, April and May
- Summer: June, July and August

Conclusion: Summer presents the highest volume of delayed flights, with delays occurring most frequently on **Mondays**, **Thursdays**, and **Sundays**, highlighting clear seasonal and weekly patterns.



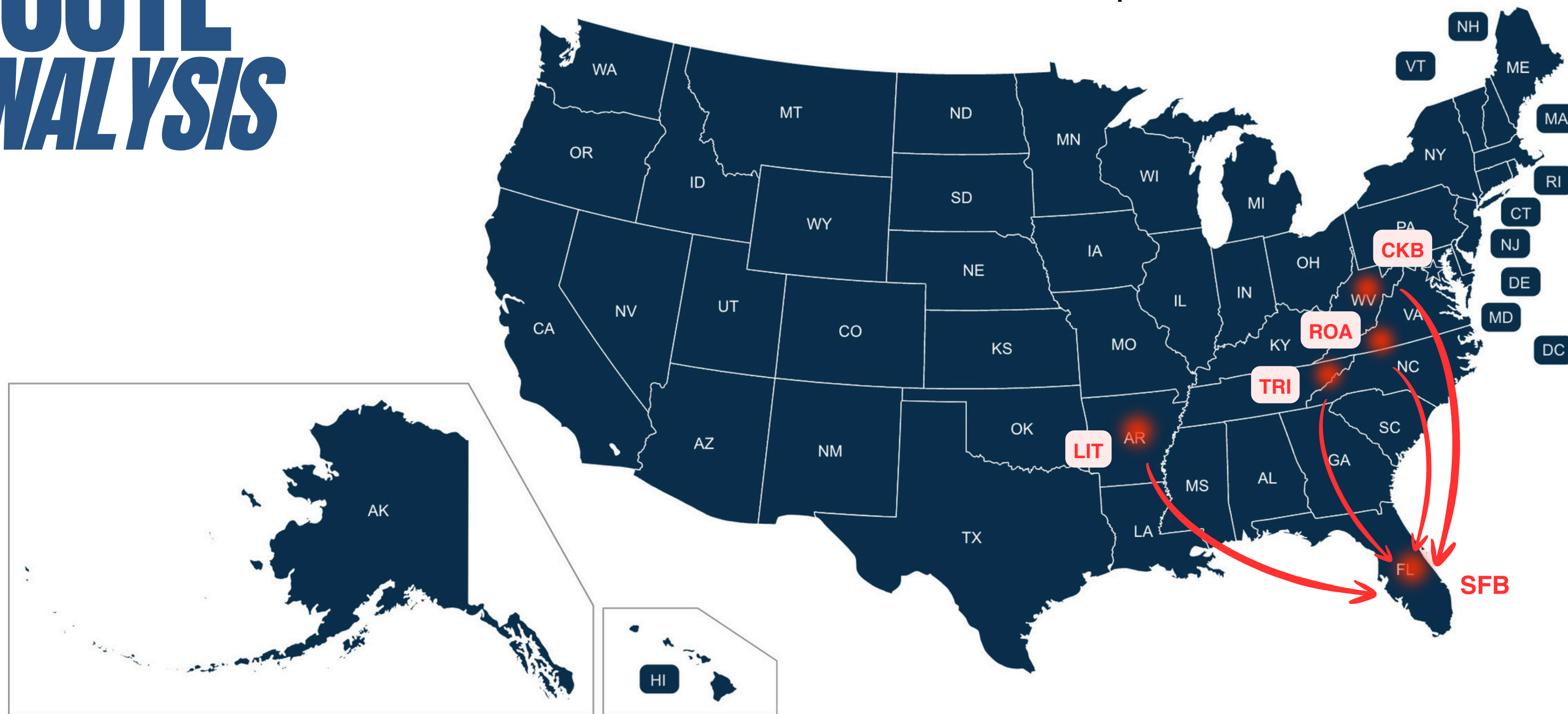
AIRLINE & AERPORT PERFORMANCE ANALYSIS



Conclusion: North Central West Virginia Airport records the highest average delays, with PSA Airlines Inc. identified as the airline most associated with delayed flights.

ROUTE ANALYSIS

The route with the **highest probability** to get **delayed** is from **Roanoke-Blacksburg Airport** to **Orlando Sanford International Airport**.



The **top four routes** with the highest probability of **delay** are all scheduled to **arrive in Orlando**, highlighting Orlando as a key destination associated with increased delay risk.

DATA SETUP *AND* FEATURE ENGINEERING

Sample of 100.000 rows

Transformed categorical features → One Hot Encoding and
numerical features → Standard Scaling

Implemented and 80%- 20% train-test split, in four different models:

- Logistic Regression
- Random Forest
- Decision Tree
- XG Boost

INITIAL XG BOOST MODEL RECALL:

- Highest recall by far
- Only model actually detecting delays

71%



IMBALANCED

Undersampling Recall:
(XG Boost)

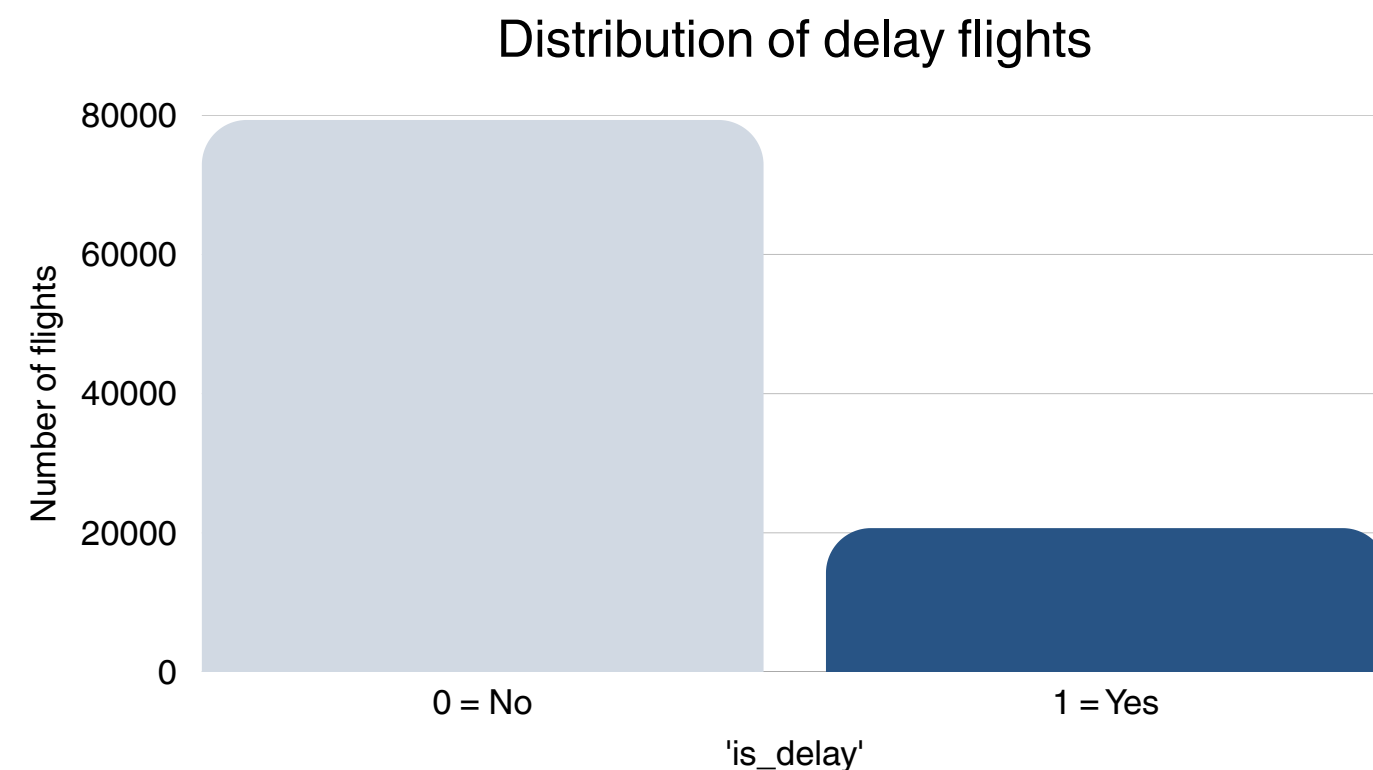


97,50%

What can we do?

- Oversampling
- Undersampling
- SMOTE

Conclusion: Overall, all models improved after addressing class imbalance; however, **XGBoost** combined with **undersampling** achieved the best performance, making it the final selected model.



Reminder: sample of 100.000 rows

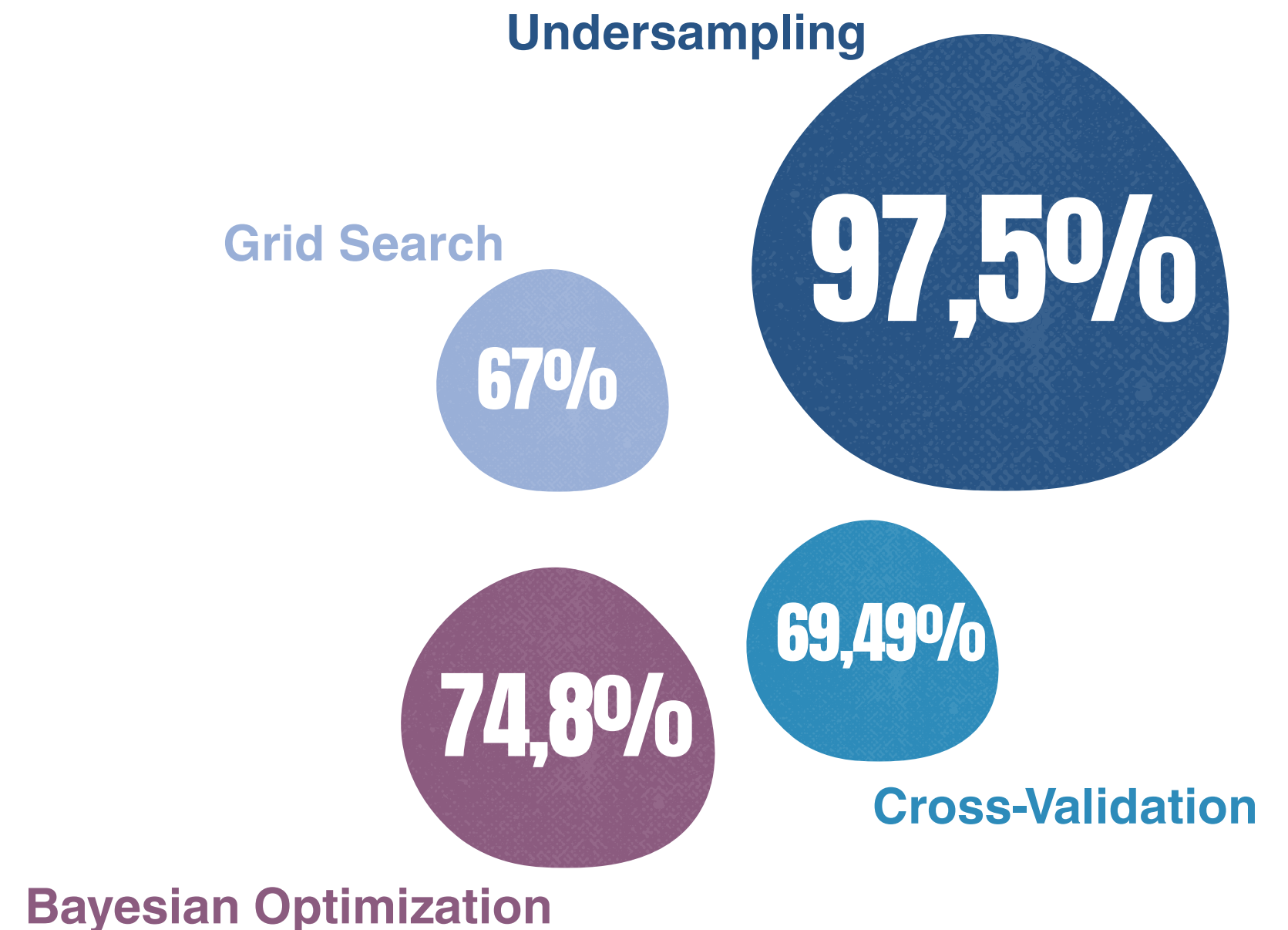
HYPERPARAMETER TUNING

Objective: Optimize model hyperparameters to maximize recall on imbalanced dataset.

Methods Tried:

- Grid Search
- Cross-Validation
- Bayesian Optimization

Conclusion: Hyperparameter tuning did not improve recall over the simple undersampling approach.



FLIGHT DELAY PREDICTION WEB

Check if your flights is going to be On-
Time or delayed:



CONCLUSIONS

KEY FINDINGS

OPERATIONAL & TEMPORAL FEATURES

FLIGHT DELAYS CAN BE PREDICTED
USING DEPARTURE TIMING,
SEASONALITY, ROUTE DYNAMICS, AND
OPERATIONAL CONDITIONS.

HIGH RECALL WITH UNDERSAMPLING

THE SELECTED MACHINE LEARNING
MODEL ACHIEVED
ITS HIGHEST RECALL AFTER APPLYING
UNDERSAMPLING,
SIGNIFICANTLY IMPROVING THE
DETECTION OF DELAYED FLIGHTS.



REAL-WORLD APPLICATION *AND* FUTURE IMPROVEMENT

In a real airline environment, this model could be integrated into scheduling and operations systems to:

- Identify flights with high delay risk before departure
- Optimize crew and aircraft allocation
- Improve gate management and turnaround planning
- Provide proactive passenger notifications

Several enhancements could further improve the performance and business value of this system:

- Incorporate real-time weather data
- Perform deeper hyperparameter optimization



THANK YOU



FLIGHT DELAY PREDICTION



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